People's Democratic Republic of Algeria
Ministry of Higher Education and Scientific Research
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Course Summary Document

Digital Law and Data Protection

Chapter 4: Algerian Legal Framework

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Introduction

Understanding the Algerian legal framework is essential for developing digital technologies, artificial intelligence (AI), cloud infrastructure, and electronic communications. This chapter synthesizes the primary legal texts, detailing the legal responsibilities of digital stakeholders without confusing legal obligations with technical cybersecurity best practices.

Key Key Legal Texts Covered:

- **Law No. 09-04 (Aug 5, 2009):** Rules on prevention and combating ICT-related offenses.
- **Law No. 18-07 (June 10, 2018):** Protection of physical persons in the processing of personal data.
- **Law No. 25-11 (July 24, 2025):** Amending and supplementing Law No. 18-07.
- **Law No. 18-04 (May 10, 2018):** General rules governing postal and electronic communications.
- **Ordinance No. 03-05 (July 19, 2003):** Copyright and neighboring rights.
- **Penal Code & Code of Criminal Procedure:** Specific provisions relating to cybercrime and automated data processing system breaches.

1. Law No. 09-04 (ICT Offenses)

1.1 Purpose and Publication Date

- Published in the Official Journal No. 47 on August 16, 2009 (dated August 5, 2009), establishing specific rules for preventing and combating ICT-related crimes.
- Key Domains: Prevention/suppression of cybercrimes, lawful surveillance, search and seizure of digital data, traffic data retention, ISP assistance to judicial authorities, and international legal cooperation.
- Must be interpreted jointly with the Penal Code, Code of Criminal Procedure, and electronic communications/privacy laws.

1.2 Official Body Designations

- **ANPDP:** National Authority for the Protection of Personal Data
- **ARPCE:** Regulatory Authority for Post and Electronic Communications
- **ISSA:** Information Systems Security Agency
- **NICC/GN:** National Institute of Criminalistics and Criminology of the National Gendarmerie
- National Penal Pole for Combating Crimes Related to Information and Communication Technologies
- **ONPLITIC:** National Authority for the Prevention of Crimes Related to Information and Communication Technologies (Law No. 09-04 mandates its uniform designation as ONPLITIC).

1.3 Main Definitions

- **Information System:** An isolated device or interconnected network of devices performing automated data processing pursuant to a program (includes servers, workstations, network devices, storage, IoT, apps, and databases).
- **ICT-Related Offense:** Offenses directly targeting systems, committed via systems, or facilitated by

ICT infrastructure.

- **Traffic Data:** Data produced by communication chains indicating origin, destination, route, time, duration, size, and service type (distinct from actual communication content).

1.4 Electronic Communications Surveillance

- Lawfully permits setting up technical devices for monitoring, real-time collection/recording of content, system searches, and data seizure under strict judicial oversight.
- Does NOT grant arbitrary, general surveillance powers to ISPs or system administrators.

1.5 Search and Seizure of Computer Data

- Authorized authorities may access systems, inspect/copy stored data, extend searches to connected systems, seal original media, and block unauthorized access while preserving data integrity.
- Forensic Best Practices: Documenting context, generating forensic images, computing cryptographic hashes (e.g., SHA-256; MD5 is discouraged due to collision risks), maintaining strict chain of custody, and working strictly on working copies.

1.6 Service Providers' Obligations

- Must assist judicial investigations by capturing/recording content data when legally ordered.
- Mandatory Traffic Data Retention: Must retain user identity, terminal data, technical parameters, timestamp/duration, and destination site details for exactly ONE YEAR (Article 11) from recording.

1.7 Example: Unauthorized Access

- Scenario: A consultant with authorized vulnerability-testing scope exploits an SQLi vulnerability to access or extract database tables.
- Legal Qualification: Constitutes fraudulent access, maintenance, or data tampering. Vulnerability discovery does not imply authorization to exploit.

1.8 Example: DDoS Attack Against a University

- Legal Qualification: Constitutes obstruction of information system functioning.
- Response Steps: Activate incident response plan, preserve logs, contact ISP/host, notify authorities, and avoid attributing attacks solely to IP addresses (IPs can be spoofed, shared, or compromised).

1.9 Example: Ransomware

- Response Steps: Isolate compromised systems, preserve logs/IOCs, execute business continuity plans, verify backup integrity, notify authorities, and assess personal data breach implications.
- Ransom Payment: Not explicitly banned by Law 09-04, but poses severe legal/financial risks (especially if funding sanctioned/terrorist entities). Requires thorough multidisciplinary assessment.

2. Law No. 18-07 (Personal Data Protection)

2.1 Scope and Purpose

- Published June 10, 2018; regulates automated and non-automated processing of personal data to protect physical persons' privacy and fundamental rights.

2.2 Personal Data

- Any information relating to an identified or identifiable physical person (e.g., name, ID number, photo, phone, email, health data, IP/MAC addresses, location, biometrics).
- IP/MAC addresses are personal data when they enable direct or indirect identification within context.

2.3 Data Processing

- Any operation performed on personal data: collection, recording, organization, storage, adaptation, retrieval, transmission, encryption, pseudonymization, erasure, or AI model training.

2.4 Data Controller

- The entity (company, university, bank, public body) that determines the purpose and essential means of data processing.
- Key Responsibilities: Purpose specification, legal basis verification, data minimization, user notification, compliance with ANPDP formalities, data security, and processor oversight.

2.5 Data Processor

- Processes data on behalf of the controller under a binding contract specifying purpose, security, sub-processing limits, audit rights, and data deletion/return.

2.6 ANPDP

- The independent National Authority overseeing data protection compliance via declarations, authorizations, inspections, and corrective sanctions.

2.7 Principles Applicable to Processing

- **Specified Purpose:** Data must be collected for specified, explicit, and legitimate purposes.
- **Proportionality & Minimization:** Only data strictly necessary for the purpose must be collected.
- **Accuracy:** Data must be accurate and kept up to date.
- **Storage Limitation:** Data must not be stored indefinitely in identifiable form; retention schedules depend on legal/statutory mandates.
- **Security & Confidentiality:** Implementation of technical/organizational controls (RBAC, MFA, encryption, network segmentation, logging, backups, intrusion detection).

2.8 Example: University Platform

- Requires defining processing purpose, restricting teacher access strictly to relevant students, retaining access logs, securing national IDs, and managing Cloud vendors.

2.9 Example: Health Data

- Highly sensitive data requiring strict Role-Based Access Control (RBAC), least privilege, strong authentication, encrypted storage, and audit trail logging.

2.10 Cross-Border Data Transfers

- Transfers abroad are subject to Articles 44 & 45. They are not absolute bans but require evaluating destination country protection levels, ANPDP authorization, and contractual safeguards.

3. Law No. 25-11 (Amendments to Data Protection)

3.1 Purpose

- Enacted July 24, 2025 (Official Journal No. 48) to amend and supplement Law No. 18-07. Focuses strictly on updated personal data protections rather than general AI legislation.

3.2 Legal Effect of the Amendment

- Must be read combinedly with Law No. 18-07, updating specific articles, formal requirements, and introducing specialized administrative mandates.

3.3 Artificial Intelligence and Personal Data

- AI model training utilizing personal data must strictly respect Law 18-07 principles: valid legal grounds, purpose limitation, minimization, security, and data subject rights.

3.4 Profiling and Automated Decision-Making

- Automated profiling (for recruitment, credit scoring, health risk) requires rigor: documentation, data quality checks, bias testing, human oversight, and contestability options.

3.5 Biometrics

- High-risk data requiring proof of strict necessity, encryption of biometric templates, non-retention of raw images, and ANPDP compliance.

3.6 Data Breach Obligations

- Data Controllers must report personal data breaches to the ANPDP within FIVE (5) DAYS of awareness. Processors must notify controllers immediately upon discovery.

4. Legal Qualification of Digital Assets

4.1 Source Code and Software

- Protected as computer programs under Copyright Ordinance 03-05 (Art. 4). Protection arises automatically upon creation without mandatory registration (ONDA deposit serves as proof).
- Author's Exclusive Rights: Reproduction, distribution, rental, adaptation, and translation (Art. 27).
- Lawful Exceptions: Legitimate users may make backup/archival copies or essential operational adjustments (Art. 52-53).

- Decompilation vs Intrusion: Reverse engineering for interoperability is governed by copyright/contract law, whereas unauthorized access/hacking falls under Penal Code Art. 394 bis.

4.2 Databases

- Protected structurally under Ordinance 03-05 (Art. 5) if original in selection/arrangement.
- Database Content: Individual contents are governed separately by personal data laws (Law 18-07/25-11), trade secret rules, or criminal statutes.

4.3 Raw Data

- Isolated raw facts/data are NOT eligible for copyright protection (Ordinance 03-05, Art. 7 excludes ideas/facts).
- Protected via personal data regulations, trade secrets, contractual confidentiality, or criminal laws against unauthorized extraction.

4.4 IP Addresses, MAC Addresses, and Logs

- Qualified as personal data whenever combined with user accounts/connection logs to identify a person.
- ISPs/Providers must retain traffic logs for 1 year (Law 09-04, Art. 11). Private companies must maintain activity registers (Law 25-11 Art. 41 bis 2 & 41 bis 3) with defined retention and purge schedules.

4.5 Content vs. Metadata

- Content (SMS body, email text, audio) is strictly protected by correspondence secrecy; real-time intercept requires explicit judicial authorization (Law 09-04 Art. 3-4).
- Metadata (timestamps, IP routes, duration) constitutes traffic data subject to mandatory retention and regulatory oversight.
- Network Admin Rules: Admins may analyze traffic solely for security within authorized scope; inspecting raw content using tools like Wireshark without authorization is illegal.

5. Key Digital Stakeholders & Responsibilities

5.1 Data Controller

- Determines purpose/means of processing. Must maintain Processing Activity Registers (Art. 41 bis 2), Automated Operation Logs (Art. 41 bis 3), perform DPIAs (Art. 45 bis 6), and report breaches within 5 days.

5.2 Data Processor

- Processes data solely on controller instructions. Must maintain processor registers, implement security, and notify controllers immediately of data breaches (Art. 45 bis 8).

5.3 Data Protection Officer (DPO)

- Mandated under Law 25-11 (Art. 41 bis). Advises controllers/staff, monitors compliance, assists with

DPIAs, and serves as primary point of contact with ANPDP.

5.4 Software Publisher

- Bound by copyright laws and contractual security terms. Intentionally introducing backdoors triggers criminal liability under Penal Code Art. 394 bis.

5.5 Cloud Host Provider

- Acts as processor or independent controller depending on service layer. Subject to data localization, cross-border rules, security mandates, and service provider retention duties under Law 09-04.

5.6 Telecom Operators and ISPs

- Regulated by ARPCE (Law 18-04). Must preserve communication secrecy, assist judicial inquiries (Law 09-04 Art. 10), and retain traffic data for 1 year (Art. 11).

5.7 Specialized Security & Forensic Bodies

- ANPDP: Personal data regulatory authority.
- ARPCE: Telecom and postal regulator.
- ISSA: Information systems security oversight.
- NICC/GN: Forensic and criminalistic digital investigations.
- National Penal Pole: Prosecution of major ICT cybercrimes.
- ONPLITIC: Coordination and prevention of ICT crimes (Law 09-04 Art. 13-14).

6. Cloud Shared Responsibility Model

Legal responsibility in Cloud computing depends on the technical service model (IaaS, PaaS, SaaS) and contractual terms:

Model	Customer Responsibilities	Provider Responsibilities
IaaS	Data, OS patching, access control, app config, legal compliance.	Physical facilities, hardware, hypervisor, physical network.
PaaS	Data, app code, user identities, access permissions.	Runtime environment, underlying OS, infrastructure maintenance.
SaaS	Data governance, user roles, security settings, compliance.	Application logic, uptime, application security, host infrastructure.

7. AI & Personal Data Lifecycle

7.1 Data Collection

- Requires defined purpose, lawful legal basis, minimization, and ANPDP registration before collecting dataset samples.

7.2 Preparation & Training

- Mandates pseudonymization, access restriction, model robustness testing, and version tracking. Pseudonymized data remains personal data if re-identification is possible (Law 25-11 Art. 3).

7.3 Profiling and Automated Decisions

- Law 25-11 (Art. 45 bis 1) restricts sole reliance on automated processing for decisions producing legal effects on persons, requiring human intervention mechanisms.

7.4 Data Protection Impact Assessment (DPIA)

- Mandatory under Law 25-11 (Art. 45 bis 6) prior to deploying high-risk processing, including AI processing sensitive health or biometric data.

7.5 Foreign API Integration Example

- A health startup using a foreign AI API must strip identifiers, verify foreign protection adequacy (Law 18-07 Art. 44-45), execute data processor agreements, conduct a DPIA, and ensure encryption.

7.6 Data Breach Management

- Requires containment, log retention, reporting to ANPDP within 5 days, notifying affected individuals if high-risk (Art. 45 bis 10), and maintaining internal breach logs.

8. Summary Table of Legal Texts & Scope

Topic / Stakeholder	Primary Legal Texts	Key Relevant Articles
Source Code & Software	Ordinance 03-05; Penal Code; Law 18-07 / 25-11	Ord. 03-05 Art. 3, 4, 27, 52, 53; Penal Code Art. 394 bis+
Databases	Ordinance 03-05; Law 18-07 / 25-11; Penal Code	Ord. 03-05 Art. 5; Law 18-07 Art. 3, 42, 44-45; Law 25-11 Art. 41 bis 2-3
Raw Data	Law 18-07 / 25-11; Ordinance 03-05; Penal Code	Law 18-07 Art. 3, 42; Law 25-11 Art. 3, 45 bis 7
IP, MAC & Logs	Law 09-04; Law 18-04; Law 18-07 / 25-11	Law 09-04 Art. 2, 10, 11; Law 25-11 Art. 41 bis 3
Communication Content	Law 09-04; Law 18-04; Penal Code	Law 09-04 Art. 3, 4, 9, 10

Data Controller	Law 18-07; Law 25-11	Law 18-07 Art. 3, 7-9, 42; Law 25-11 Art. 41 bis 2, 45 bis 6, 8
Data Processor	Law 18-07; Law 25-11	Law 18-07 Art. 3, 41-42; Law 25-11 Art. 41 bis 2-3, 45 bis 8
DPO	Law 25-11	Law 25-11 Art. 41 bis, 41 bis 1
Software Publisher	Ordinance 03-05; Penal Code; Law 18-07 / 25-11	Ord. 03-05 Art. 4, 27, 52-53; Penal Code Art. 394 bis+
Cloud Host Provider	Law 09-04; Law 18-04; Law 18-07 / 25-11	Law 09-04 Art. 2, 10, 11; Law 18-07 Art. 42, 44-45; Law 25-11 Art. 45 bis 7
Telecom & ISPs	Law 18-04; Law 09-04; Law 18-07 / 25-11	Law 09-04 Art. 10-12; Law 18-04 general rules
ANPDP	Law 18-07; Law 25-11	Law 25-11 Art. 27 bis, 41 bis, 45 bis 6-13
ARPCE	Law 18-04	Law 18-04 regulatory framework
ONPLITIC	Law 09-04	Law 09-04 Art. 13, 14